

0.4 Functions

Functions are the mathematical way to describe *processes* or *transformations*—rules that take an input and produce an output. In linear algebra, the central objects of study are **linear maps** (also called linear transformations), which are functions between vector spaces that respect the vector space structure. Understanding functions in general—especially their composition and inverses—is crucial because:

- **Matrix multiplication** corresponds to **composition** of linear maps.
- An **invertible matrix** corresponds to a **bijective** linear map.
- Many proofs involve showing a linear map is injective, surjective, or bijective.

0.4.1 Definition and Basic Properties

Definition 0.4.1 — Function

A **function** f from a set A to a set B , denoted $f : A \rightarrow B$, is a rule that assigns to each element $a \in A$ exactly one element $f(a) \in B$. The set A is called the **domain** and B is called the **codomain**.

Definition 0.4.2 — Injective, Surjective, Bijective

Let $f : A \rightarrow B$ be a function.

- f is **injective** (one-to-one) if $f(a_1) = f(a_2)$ implies $a_1 = a_2$.
- f is **surjective** (onto) if for every $b \in B$, there exists $a \in A$ such that $f(a) = b$.
- f is **bijective** if it is both injective and surjective

Example. (Injective, Surjective, and Bijective Functions)

Consider the following functions:

1. $f : \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = 2x + 1$ is **bijective**.
 - Injective: If $2x_1 + 1 = 2x_2 + 1$, then $x_1 = x_2$.
 - Surjective: For any $y \in \mathbb{R}$, we have $f\left(\frac{y-1}{2}\right) = y$.
2. $g : \mathbb{R} \rightarrow \mathbb{R}$, $g(x) = x^2$ is **neither injective nor surjective**.
 - Not injective: $g(1) = g(-1) = 1$, but $1 \neq -1$.
 - Not surjective: There is no $x \in \mathbb{R}$ such that $g(x) = -1$.
3. $h : \mathbb{R} \rightarrow [0, \infty)$, $h(x) = x^2$ is **surjective but not injective**.
 - Not injective: $h(1) = h(-1) = 1$.
 - Surjective: For any $y \geq 0$, we have $h(\sqrt{y}) = y$.
4. $k : [0, \infty) \rightarrow \mathbb{R}$, $k(x) = x^2$ is **injective but not surjective**.
 - Injective: If $x_1^2 = x_2^2$ with $x_1, x_2 \geq 0$, then $x_1 = x_2$.
 - Not surjective: There is no $x \geq 0$ such that $k(x) = -1$.

0.4.2 Function Composition

When we apply one function after another, we obtain a **composition**. This concept is fundamental: matrix multiplication is precisely the composition of the corresponding linear maps.

Definition 0.4.3 — Function Composition

Let $f : A \rightarrow B$ and $g : B \rightarrow C$ be functions. The **composition** of g with f , denoted $g \circ f$, is the function $g \circ f : A \rightarrow C$ defined by:

$$(g \circ f)(a) = g(f(a)) \quad \text{for all } a \in A$$

We read $g \circ f$ as “ g composed with f ” or “ g after f .”

Remark. The notation $g \circ f$ means “first apply f , then apply g .” The order matters and can be confusing—the function written on the right is applied first. This mirrors how matrix multiplication works: in $\mathbf{A}\mathbf{B}\mathbf{x}$, we first multiply $\mathbf{B}\mathbf{x}$, then multiply by $\mathbf{A}$.

Example. (Function Composition)

Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be defined by $f(x) = x^2$ and $g : \mathbb{R} \rightarrow \mathbb{R}$ by $g(x) = x + 1$. We compute both compositions:

- $(g \circ f)(x) = g(f(x)) = g(x^2) = x^2 + 1$
- $(f \circ g)(x) = f(g(x)) = f(x + 1) = (x + 1)^2$

Note that $g \circ f \neq f \circ g$ in general—composition is **not commutative**.

Theorem 0.4.1 — Associativity of Composition

Let $f : A \rightarrow B$, $g : B \rightarrow C$, and $h : C \rightarrow D$ be functions. Then:

$$(h \circ g) \circ f = h \circ (g \circ f)$$

That is, function composition is **associative**.

Proof. For any $a \in A$:

$$((h \circ g) \circ f)(a) = (h \circ g)(f(a)) = h(g(f(a)))$$

and

$$(h \circ (g \circ f))(a) = h((g \circ f)(a)) = h(g(f(a)))$$

Since both expressions equal $h(g(f(a)))$ for all $a \in A$, the compositions are equal. ■

Theorem 0.4.2 — Composition Preserves Injectivity and Surjectivity

Let $f : A \rightarrow B$ and $g : B \rightarrow C$ be functions.

1. If f and g are both injective, then $g \circ f$ is injective.
2. If f and g are both surjective, then $g \circ f$ is surjective.
3. If f and g are both bijective, then $g \circ f$ is bijective.

Proof. We prove (1). Suppose f and g are injective and $(g \circ f)(a_1) = (g \circ f)(a_2)$. Then $g(f(a_1)) = g(f(a_2))$. Since g is injective, $f(a_1) = f(a_2)$. Since f is injective, $a_1 = a_2$. Thus $g \circ f$ is injective.

The proof of (2) and (3) are similar exercises. ■

0.4.3 Identity Function

Definition 0.4.4 — Identity Function

For any set A , the **identity function** $\text{id}_A : A \rightarrow A$ is defined by:

$$\text{id}_A(a) = a \quad \text{for all } a \in A$$

Remark. The identity function satisfies $f \circ \text{id}_A = f$ for any $f : A \rightarrow B$, and $\text{id}_B \circ f = f$. It is the “do nothing” function. In matrix terms, the identity matrix $\mathbf{I}_n$ represents the identity function on F^n .

0.4.4 Inverse Functions

A bijective function can be “undone”—this leads to the concept of an inverse function. Understanding inverses is essential because **invertible matrices** are precisely those representing bijective linear maps.

Definition 0.4.5 — Inverse Function

Let $f : A \rightarrow B$ be a function. A function $g : B \rightarrow A$ is called an **inverse** of f if:

$$g \circ f = \text{id}_A \quad \text{and} \quad f \circ g = \text{id}_B$$

If such a g exists, we say f is **invertible** and write $g = f^{-1}$.

Theorem 0.4.3 — Bijective if and only if Invertible

A function $f : A \rightarrow B$ is bijective if and only if it has an inverse $f^{-1} : B \rightarrow A$.

Moreover, if the inverse exists, it is unique.

Proof. ($\Rightarrow$) Suppose f is bijective. For each $b \in B$, since f is surjective, there exists some $a \in A$ with $f(a) = b$. Since f is injective, this a is unique. Define $g(b) = a$. We verify that g is an inverse:

- $(g \circ f)(a) = g(f(a)) = a$, so $g \circ f = \text{id}_A$.
- $(f \circ g)(b) = f(g(b)) = f(a) = b$, so $f \circ g = \text{id}_B$.

Thus $g = f^{-1}$ is an inverse of f .

($\Leftarrow$) Suppose f has an inverse g . We show f is bijective:

- *Injective:* If $f(a_1) = f(a_2)$, then $a_1 = g(f(a_1)) = g(f(a_2)) = a_2$.
- *Surjective:* For any $b \in B$, we have $f(g(b)) = b$, so b is in the image of f .

Uniqueness: If g and h are both inverses of f , then:

$$g = g \circ \text{id}_B = g \circ (f \circ h) = (g \circ f) \circ h = \text{id}_A \circ h = h$$

■

Example. (Inverse Functions)

The following examples illustrate inverse functions and when they do or do not exist:

1. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be defined by $f(x) = 2x + 3$. This is bijective, and its inverse is $f^{-1}(y) = \frac{y-3}{2}$.
Verification: $f(f^{-1}(y)) = 2 \cdot \frac{y-3}{2} + 3 = y$ and $f^{-1}(f(x)) = \frac{(2x+3)-3}{2} = x$.
2. Let $g : \mathbb{R} \rightarrow \mathbb{R}$ be defined by $g(x) = x^2$. This is not bijective (not injective), so g^{-1} does not exist as a function $\mathbb{R} \rightarrow \mathbb{R}$.
3. Let $h : [0, \infty) \rightarrow [0, \infty)$ be defined by $h(x) = x^2$. With restricted domain and codomain, h is bijective, and $h^{-1}(y) = \sqrt{y}$.

Theorem 0.4.4 — Inverse of a Composition

If $f : A \rightarrow B$ and $g : B \rightarrow C$ are both bijective, then $g \circ f$ is bijective and:

$$(g \circ f)^{-1} = f^{-1} \circ g^{-1}$$

Proof. We verify the inverse property:

$$(f^{-1} \circ g^{-1}) \circ (g \circ f) = f^{-1} \circ (g^{-1} \circ g) \circ f = f^{-1} \circ \text{id}_B \circ f = f^{-1} \circ f = \text{id}_A$$

Similarly, $(g \circ f) \circ (f^{-1} \circ g^{-1}) = \text{id}_C$.

■